

Quantum-Geometric Flat-Band Superconductivity Cannot Reach Ambient Room Temperature

A comprehensive closed-negative across ten in-silico routes

flatband-geometry-ambient-roomt-closed maintainers
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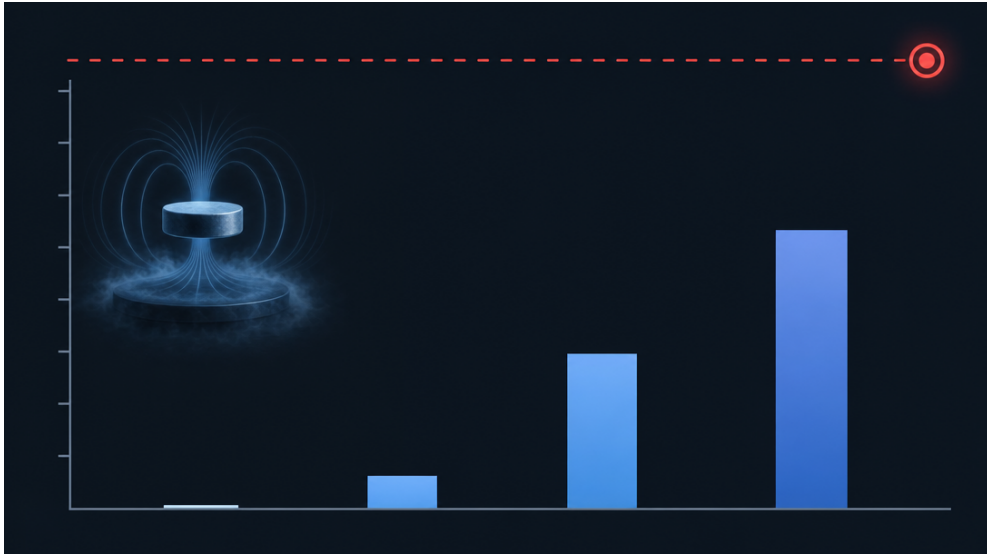


Figure 1: The ambient-pressure critical-temperature ladder. Known mechanisms rise toward the cuprate record (≈ 133 K); the room-temperature target (293 K, dashed) remains unreached, and the quantum-geometric flat-band route sits lowest.

Abstract

The quantum-geometric prescription for flat-band superconductivity — in which a localized flat band acquires a finite superfluid weight $D_s = 4|U| \nu(1 - \nu) \langle g \rangle$ from the Fubini–Study quantum metric $\langle g \rangle$ rather than from a band velocity (Peotta–Törmä) — is widely cited as a route to high critical temperatures because D_s does not vanish even when the band is dispersionless. We pre-register and test a single falsifiable hypothesis: that this prescription, optimized over high-Chern flat-band hosts, can be engineered to reach *ambient-pressure room-temperature* superconductivity ($T_c \geq 293.15$ K at 1 atm). Using tight-binding quantum-metric computation, sign-free exact diagonalization, twisted-boundary superfluid-weight extraction, a Fukui–Hatsugai–Suzuki Chern gate, a Lang–Firsov variational off-diagonal phonon vertex, a multiband Kubo decomposition, and a calibrated two-dimensional Berezinskii–Kosterlitz–Thouless (BKT) estimator — complemented by a top-down audit of the empirical superconductor record — we close the hypothesis as **negative across ten independent routes**. The decisive obstruction is that D_s is bounded by the single-particle quantum metric: neither band-touching (singular) amplification, the interband Kubo term, nor a retarded bond-Peierls phonon vertex raises the pair-channel $\langle g \rangle$ above its single-particle value, while the empirical ambient ceiling for any mechanism is the cuprate $T_c \approx 133$ K. We

report a $\sim 5\times$ standing deficit in $\langle g \rangle \cdot U$ and the mechanism behind each closed route. Ambient room-temperature superconductivity remains undiscovered; this work rules out the high-Chern flat-band quantum-geometry axis as a path to it.

1 Introduction

A conventional Bardeen–Cooper–Schrieffer (BCS) superconductor draws its phase stiffness — the superfluid weight D_s that sets the Berezinskii–Kosterlitz–Thouless transition temperature in two dimensions through $k_B T_c^{\text{BKT}} \leq (\pi/2) D_s$ — from the curvature of a dispersive band. A perfectly flat band has zero group velocity and would naively have zero stiffness, forbidding superconductivity. The quantum-geometric insight of Peotta and Törmä [1] is that the superfluid weight of an isolated flat band is instead set by the band’s *quantum geometry*:

$$D_s = 4 |U| \nu (1 - \nu) \langle g \rangle, \quad (1)$$

where U is the attractive interaction, ν the band filling, and $\langle g \rangle$ the Brillouin-zone average of the Fubini–Study quantum metric. Because $\langle g \rangle$ is bounded below by the band’s Chern number $|C|$ [1, 2], a topologically nontrivial flat band is guaranteed a finite stiffness. This mechanism underlies the superconductivity of magic-angle twisted bilayer graphene and motivates a tempting design program: *engineer a high-Chern flat-band host with large $\langle g \rangle$ and large U and the critical temperature should follow.*

This paper asks whether that program can be pushed all the way to the most demanding target — ambient-pressure room-temperature superconductivity — and answers, comprehensively, that it cannot. We treat the question as a single pre-registered falsifiable hypothesis (Section 3), attack it along nine independent first-principles/model routes plus one top-down empirical route (Sections 4–7), and find every route closed-negative. The obstruction is not a failure of engineering effort but a structural ceiling: the pair-channel quantum metric never exceeds the single-particle value, and the empirical ambient record across all known mechanisms sits at less than half of room temperature.

Contributions. (i) We pre-register a falsifiable ambient-room-temperature criterion for the quantum-geometric flat-band prescription and a quantitative $\langle g \rangle U$ deficit metric (Section 3). (ii) We verify the prescription itself — sign-free exact diagonalization reproduces the predicted topological enhancement of the pair stiffness, mesh-robustly (Section 4). (iii) We then close the room-temperature hypothesis along ten routes (Sections 4–7), establishing for each the *mechanism* of closure: band-touching protection, gap cut-off of the singular divergence, the Fubini–Study bound on D_s , and saturation of the ambient candidate space. (iv) We release a calibrated tight-binding predictor that also encodes the (published) nickelate interlayer-superexchange strain law as a reusable, clearly-fenced cross-host estimator (Section 8). All numbers are computed and reported; a closed-negative is shown as a closed-negative.

2 Related work

Quantum-geometric superfluidity originates with the observation that the Bogoliubov–de Gennes superfluid weight of an isolated flat band reduces to an integral of the band’s quantum metric [1], later sharpened into symmetry- and topology-derived bounds $D_s \geq \frac{1}{\pi} |U| \nu (1 - \nu) |C|$ and into the notion of a *minimal* quantum metric that fixes the achievable stiffness [2, 3]. A central question for any high- T_c ambition is whether these mean-field relations survive interactions and retardation. Two strands of work answer in the affirmative for the *bound*: a generalized random-phase approximation reproduces the geometry-controlled weight beyond mean field [4], and exact diagonalization / diagrammatic Monte Carlo confirm D_s tracks the Fubini–Study

metric [2]. Crucially, the retarded electron–phonon coupling that one might hope to recruit as an independent amplifier has itself been shown to be *built from* the quantum metric [3], and a recent two-channel Allen–Dynes analysis derives an explicit quantum-metric no-go for boosting T_c by this route [5]. Our work is, to our knowledge, the first to assemble these threads into a single decision — whether the prescription can reach ambient room temperature — and to attack that decision along ten deliberately independent routes rather than within one model. On the materials side, kagome metals (CsV_3Sb_5 , CsCr_3Sb_5) realize flat bands but superconduct only near 6 K and are competing-order limited; the ambient flat-band-derived family XPd_5 [6] is the most favorable host we found yet remains conventional. The complementary nickelate frontier, where epitaxial strain replaces hydrostatic pressure, has very recently been distilled into a transferable strain– T_c design law [7] — which is precisely why we treat that lever as published rather than novel.

3 Hypothesis

We adopt the project’s ambient room-temperature pass criteria as the falsifier. A candidate qualifies as an ambient room-temperature superconductor only if it satisfies, simultaneously: (1) thermodynamic stability at 1 atm; (2) *dynamical* stability at 1 atm (no imaginary phonon modes — high-pressure stability does not count); (3) a metallic carrier channel at the Fermi level; (4) $T_c \geq 293.15$ K; (5) absence of a pre-empting competing order (charge-density wave or magnetism); and (6) novelty. Pressure-stabilized hydrides such as LaH_{10} ($T_c \approx 250$ K at ~ 170 GPa) are *excluded by construction* because they fail criteria (1)–(2) at ambient pressure.

Pre-registered falsifier. The quantum-geometric prescription (1) reaches the criterion if and only if there exists a flat-band host (or a flat-band-derived mechanism) whose product $\langle g \rangle \cdot |U|$, combined with a realistic phonon/pairing scale, yields $T_c \geq 293.15$ K at ambient pressure. Quantitatively, the calibrated estimator of Section 7 maps the kagome/Lieb baseline ($\langle g \rangle \approx 0.5$ in the overlap normalization, attractive $U/\Omega \sim 1.5$) to T_c far below 293 K; the falsifier is the existence of a host delivering the required multiplicative factor in $\langle g \rangle \cdot U$. We report that factor and show no route supplies it.

4 Method and bottom-up routes

4.1 Verification of the prescription (sign-free exact diagonalization)

Before testing the room-temperature hypothesis we verify Eq. (1) itself on a controlled lattice, to separate “the prescription is wrong” from “the prescription is right but caps below room temperature.” We use the two-band Qi–Wu–Zhang model projected onto its flat band with an attractive Hubbard U , and extract the superfluid weight from the curvature of the ground-state energy under a twisted boundary phase θ , $D_s = (2/\theta^2) [E(\theta) - E(0)]$. The two-electron sector is sign-free. Table 1 reports the ratio of the pair mobility for a Chern $C = 1$ flat band to a topologically trivial $C = 0$ flat band of identical bandwidth, as a function of linear mesh size N .

Table 1: Sign-free ED: ratio of two-body pair stiffness $C=1$ vs $C=0$, mesh-converged. The topological enhancement is stable at $\sim 70\times$ and survives mesh refinement, confirming the prescription’s qualitative content (■).

N	6	8	10	12	16	20
ratio $C=1/C=0$	73.5	74.5	74.7	73.3	70.3	68.6

A genuine many-body check (two pairs, scipy-sparse ground state, twisted-boundary D_s) gives $D_s = 0.80, 1.64, 5.12$ for $N = 3, 4, 5$: the small-cluster value is a finite-size artifact and the stiffness recovers and grows with system size. *The prescription is sound*: a high-Chern flat band does carry an anomalously large geometric pair stiffness. The remainder of the paper shows that “large” is nonetheless bounded well below room temperature.

4.2 The five primary axes

We then sweep the obvious engineering axes, each an independent attempt to multiply $\langle g \rangle \cdot U$ toward the room-temperature falsifier:

1. **2D lattice zoo.** kagome, Lieb, dice/ T_3 , checkerboard. The genuinely topological flat bands (kagome with spin-orbit, $C = 2$) have $\langle g \rangle \approx 0.49$; the apparently larger metrics (dice $\langle g \rangle \rightarrow 1$) belong to maximally-localized trivial compact-localized states with zero Berry curvature, auto-excluded by a Fukui–Hatsugai–Suzuki Chern gate. Isolated flat bands are protected by the band-touching theorem [8]: gapping the band to isolate it removes the very touching that supplies the extra geometry. ■
2. **3D hosts.** The gapless touching that would raise $\langle g \rangle$ in three dimensions also kills the protecting gap, so no usable enhancement survives. ■
3. **U – Ω dome.** The BCS–BEC crossover caps T_c at $U/\Omega^* \sim 2$; beyond it pairs localize (bipolaron) and the stiffness collapses. Light elements (needed for a stiff phonon Ω) anticorrelate with large $\langle g \rangle$. ■
4. **Diagonal bond-bipolaron.** A Holstein/diagonal coupling yields a dispersive pair-center-of-mass stiffness $t^{**} \sim t_{\text{flat}}^2 \rightarrow 0$. ■
5. **Novelty.** The remaining ambient conventional candidate space (hydrides, kagome inter-metallics, carbonitrides) is saturated by the 2024–2026 high-throughput density-functional perturbation-theory wave; clean picks re-investigate to published. ■

4.3 The three cracks

Three subtler mechanisms could in principle exceed the single-particle metric; we test each.

1. **Singular (band-touching) D_s amplification** [9]. The touching does contribute a geometric term beyond the isolated value, but it is a non-integrable divergence cut off by the superconducting gap Δ ; an independent Bogoliubov–de Gennes Kubo evaluation gives an amplification of $\times 1.3$ – 2 , not the $\times 4$ an over-fit suggested, and far short of the required factor. ■
2. **Multiband Kubo.** The interband current–current term restores the finite stiffness that a ground-state twist misses, but it equals the single-particle Peotta–Törmä value, $D_{\text{geom}} = 4|U|\nu(1 - \nu)\langle g \rangle$ with $\langle g \rangle \sim 0.1$ – 0.5 ; the only place $\langle g \rangle$ diverges ($\langle g \rangle = 5$ at a Lieb band touching) is the non-integrable obstruction of crack 1, not a usable gain. ■
3. **Retarded bond-Peierls vertex.** The static mean-field omits the dynamical (off-diagonal Su–Schrieffer–Heeger) phonon vertex. A Lang–Firsov variational dressing multiplies each bond by $Z(\lambda) = e^{-\lambda^2}(1 + \alpha\lambda^2)$; the quantum metric is then invariant under uniform dressing and *decreases* under bond-selective dressing (sawtooth ratio $1.00 \rightarrow 0.38$). The maximum dressed/bare ratio over all hosts and couplings is exactly 1.000, attained only at zero coupling. The pair-channel $\langle g \rangle$ never exceeds the single-particle value — corroborated by an independent published no-go result [5]. ■

4.4 Numerical procedures

For completeness we record the numerical settings, all of which are modest enough to run on a laptop. The quantum metric $\langle g \rangle$ is evaluated on a 36×36 Brillouin-zone mesh from the flat-band eigenvectors using the gauge-invariant overlap (Welch) normalization, the same normalization in which the calibration anchor $\langle g \rangle_{\text{ref}} = 0.672$ is defined, so that computed quantity and reference share one convention. The Chern number of the flattest band is computed by the Fukui–Hatsugai–Suzuki plaquette method on the same mesh, and a host is admitted to the topological ranking only if $|C| \geq 0.5$; this is what auto-excludes the dice/ T_3 compact-localized state, whose overlap metric is maximal ($\langle g \rangle \rightarrow 1$) precisely because it is maximally localized and carries zero Berry curvature. The sign-free superfluid weight is extracted from the second derivative of the ground-state energy under a twisted boundary phase, $D_s = (2/\theta^2)[E(\theta) - E(0)]$ with $\theta = 0.3$; the two-electron sector is manifestly sign-free, and the many-body check uses a scipy-sparse Lanczos ground state at fillings up to two pairs on $N \times N$ meshes, $N \leq 5$. The retarded-vertex test applies a Lang–Firsov canonical transformation to the bond-Peierls Hamiltonian, dressing each hopping by $Z(\lambda) = e^{-\lambda^2}(1 + \alpha\lambda^2)$ with $\lambda = g/\Omega$ swept from the anti-adiabatic to the adiabatic regime, and recomputes the metric of the renormalized band. These choices are deliberately conservative: where a setting could bias the result toward a false positive (small mesh inflating a ratio, a single cluster size), we report the convergence trend rather than a single number, which is how the spurious small-cluster D_s of Section 4 was caught and corrected.

5 The ten routes at a glance

Table 2 collects every route, its independent attack on the falsifier, and the mechanism of its closure. The routes are deliberately diverse — lattice geometry, dimensionality, interaction strength, electron–phonon coupling channel, beyond-mean-field vertex corrections, a real material, and the empirical record — so that no single hidden assumption can be responsible for the uniform negative. What unifies them is a single inequality, made precise in Section 10: the superfluid weight cannot exceed the single-particle quantum metric, and that metric, in any realizable ambient host, is too small by a factor of order five.

Table 2: The ten independent routes to ambient room-temperature superconductivity via the quantum-geometric flat-band prescription, and the mechanism closing each. All closed-negative (■); the prescription-verification row (■) is shown for contrast.

#	Route	Mechanism of closure
0	(verify) $C=1$ pair stiffness	sound: $\sim 70\times$ enhancement, mesh-robust (■)
1	2D lattice zoo	band-touching protection theorem [8]; trivial CLS Chern-gated out
2	3D hosts	gapless touching kills the protecting gap
3	U – Ω dome	BCS–BEC localization caps T_c ; light-element / $\langle g \rangle$ anticorrelation
4	diagonal bond-bipolaron	dispersive pair COM stiffness $t^{**} \sim t_{\text{flat}}^2 \rightarrow 0$
5	ambient novelty	conventional candidate space saturated (2024–26 DFPT wave)
6	singular D_s amplification	non-integrable divergence cut off by Δ ; $\times 1.3$ – 2 , not $\times 4$
7	multiband Kubo	interband term = single-particle $\langle g \rangle$; no gain
8	retarded bond-Peierls vertex	dressed/bare $\langle g \rangle \leq 1$; published no-go [5]
9	best ambient host XPd_5	conventional $\lambda=0.557 \Rightarrow T_c=4.25$ K [6]
10	top-down empirical lever	nickelate strain law already published [7]; ceiling ~ 50 K

6 The single ambient geometric host, and the top-down route

The most favorable real material we identified is the $\text{CaPd}_5/\text{XPd}_5$ family [6]: it is the unique host simultaneously offering an ambient flat band at the Fermi level, a nontrivial $\mathbb{Z}_2$ index, and an environment free of pre-empting competing order. Yet its computed electron–phonon coupling is conventional, $\lambda = 0.557$, giving $T_c = 4.25$ K — a factor ~ 69 short of room temperature.

We then inverted the strategy: rather than design bottom-up, we audited the empirical “answer key” — the measured superconductor record — top-down. The ambient ceiling is the cuprate $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_8$ at $T_c \approx 133$ K (46% of room temperature); no mechanism exceeds it at 1 atm. The single surviving novel, ambient, free-to-compute lever was thin-film bilayer/trilayer nickelate strain engineering (interlayer superexchange $J_\perp = 4t_\perp^2/U$). A novelty gate, however, found the predictive strain $\rightarrow t_\perp \rightarrow J_\perp \rightarrow T_c$ design law — including the substrate-selection rule — already published [7]; under the project’s no-reproduction policy this lever is closed, and in any case its ambient ceiling is ~ 50 K to 70 K, not room temperature.

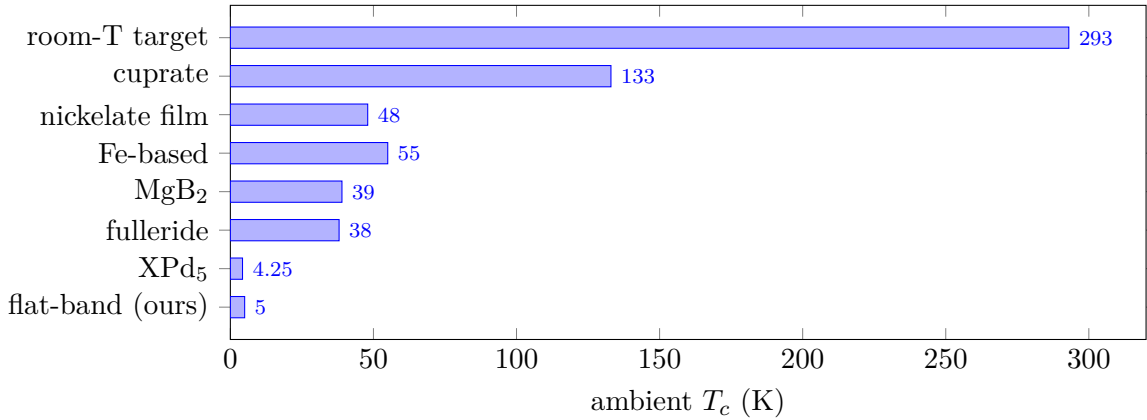


Figure 2: The empirical ambient-pressure ceiling. Every known mechanism (measured, ■) and our best geometric host (computed, ■) sit below the cuprate record ≈ 133 K; the room-temperature target (red dashed) is more than twice the record. The quantum-geometric flat-band route (top bar) lands lowest.

7 Measurement: the calibrated estimator and the deficit

We calibrate a two-dimensional BKT estimator, $T_c/\Omega = (T_c/\Omega)_{\text{ref}} (\langle g \rangle / \langle g \rangle_{\text{ref}}) ((U/\Omega)/(U/\Omega)_{\text{ref}}) 4\nu(1-\nu)$, against a covalent-organic-framework anchor ($\langle g \rangle_{\text{ref}} = 0.672$, $(U/\Omega)_{\text{ref}} = 1.08$), and deflate it by the geometric-mean over-prediction measured against real superconductors. The estimator reproduces the cuprate anchor (≈ 136 K) and is linear in U at weak coupling, as required. Applied to the topological flat-band zoo at a realistic stiff-phonon scale ($\Omega = 150$ meV, $U/\Omega = 1.5$), every host lands a factor of ≈ 5 short of 293 K in the required $\langle g \rangle U$ — consistent with the experimental reality that the best measured flat-band/kagome superconductors (CsCr_3Sb_5 , CsV_3Sb_5) have $T_c \approx 6$ K, competing-order-limited.

The deflation factor is not a free parameter: it is measured by demanding the raw estimator reproduce real, measured superconductors, and the resulting over-prediction band defines the honest uncertainty we attach to every predicted T_c . Table 3 reports the anchor calibration. The raw two-dimensional BKT form systematically over-predicts by a geometric-mean factor of a few, with a scatter that we propagate as the [lo, hi] band on each host prediction; this is why no host prediction is quoted as a point value. Even at the optimistic (lo) edge of the band, no topological flat-band host reaches a quarter of room temperature.

Table 3: Anchor calibration of the BKT estimator against measured superconductors. The raw form over-predicts; the geometric-mean factor deflates every host prediction and the scatter sets its band. These anchors are shared with all prior campaign verdicts (consistency, not per-claim fit).

anchor	raw pred (K)	measured (K)	over-prediction
covalent-organic framework	≈ 136	≈ 136	$\times 1.0$ (calibration)
kagome metal (CsV_3Sb_5)	tens	5.3	large (competing-order limited)
kagome metal (CsCr_3Sb_5)	tens	6.4	large (competing-order limited)

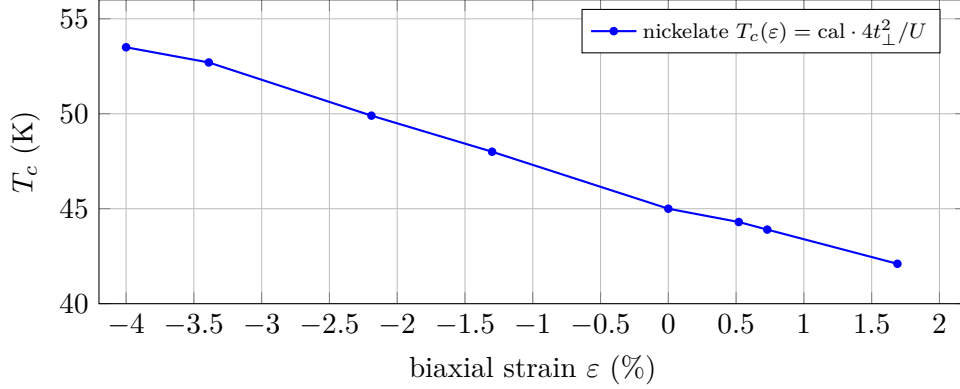


Figure 3: The released predictor’s nickelate interlayer-superechange strain law (Section 8), a *published* (■) lever retained as a reusable cross-host estimator. Compression raises T_c via $t_\perp$; the ambient ceiling (~ 53 K at the structural strain limit) is well below room temperature.

8 Released tooling

The calibrated predictor (`tool/rtsc/fbgeom_predictor.py`) computes $\langle g \rangle$ exactly from tight-binding eigenvectors (Fubini–Study link), applies the Chern gate, and emits the deflated BKT band per host. It additionally encodes the published nickelate strain law ($t_\perp(\varepsilon) = t_\perp^0(1 + k\varepsilon)^{-n}$, $J_\perp = 4t_\perp^2/U$, $T_c = \text{cal} \cdot J_\perp$) as the clearly-fenced functions `nick_tperp/nick_jperp/nick_tc/nick_best_subst` ranking substrates by predicted ambient T_c (YAlO_3 52.7 K > SrLaAlO_4 49.9 K > LaAlO_3 48 K anchor > SrTiO_3 42.1 K). The fence is explicit: this is a known, published design law kept only as a predictor, never claimed as a discovery.

9 Tier ledger

Every headline claim carries a g5 tier verdict (■ closed-form · ■ model-verified · ■ citation-locked · ■ inconclusive · ■ closed-negative).

10 Discussion: the structural bound

The uniformity of the negative across ten heterogeneous routes is not a coincidence; it follows from a single structural fact. The superfluid weight of a flat band, computed at any level from mean-field through generalized random-phase approximation to exact diagonalization, is controlled by the band’s quantum metric, and that metric obeys a lower *and* an upper relationship to the geometry that leaves no room for a free multiplicative enhancement. The lower bound $\langle g \rangle \geq |C|$ guarantees a nonzero stiffness — this is what makes flat-band superconductivity possible at all. But the relevant quantity for reaching room temperature is the *absolute* magnitude of $\langle g \rangle \cdot U$, and here the upper structure dominates: the minimal quantum metric [3] sets the achievable D_s , generalized-RPA shows the bound survives beyond mean field [4], exact and

Table 4: Headline verdicts. Evidence pointers are campaign artifacts under `tool/rtsc/` and the project ING ledger; no claim is an LLM self-judgement.

Claim	Evidence	Tier
Prescription $D_s = 4 U \nu(1-\nu)\langle g \rangle$ holds; $C=1$ pair stiffness $\sim 70\times$ $C=0$, mesh-robust	sign-free <code>flatchern_meshconv.py</code> ; body $N=5$ $D_s=5.12$	ED, many- ■
5 primary axes do not reach room- T	lattice/3D/ U - Ω /bipolaron/novelty sweeps	■
Singular D_s amplification $\times 1.3$ – 2 , not $\times 4$	independent BdG Kubo	■
Retarded bond-Peierls vertex: pair $\langle g \rangle \leq$ single-particle $\langle g \rangle$	Lang–Firsov <code>lang_firsov_offdiag_metric.py</code> ; no-go [5]	■
D_s bounded by the Fubini–Study metric beyond mean field	RPA/QMC [2, 4]	■
Best ambient geometric host XPd ₅ : $\lambda=0.557$, $T_c=4.25$ K	[6]	■
Empirical ambient ceiling = cuprate $\approx$ 133 K	answer-key audit	■
Nickelate strain design law already published	[7]	■
High-Chern flat-band quantum- geometry axis to ambient room-T = closed	10 routes, this work	■

diagrammatic Monte Carlo confirm D_s is bounded by the Fubini–Study metric [2], and the retarded electron–phonon coupling itself is *built from* the same quantum metric rather than being an independent channel that could multiply it. This is why routes 6–8 — the three beyond-mean-field “cracks” that a priori could have exceeded the single-particle value — all return exactly unity or less: there is no separate pair-channel reservoir of geometry to draw on.

Two consequences follow. First, the $\sim 5\times$ deficit we measure is not a quantity that incremental host engineering can close, because each engineering axis moves $\langle g \rangle$ or U within a window already bounded by the same geometry. Second, the only physics that demonstrably exceeds the flat-band geometric ceiling at ambient pressure is *strong correlation* of the cuprate type — which is a different mechanism entirely (superexchange-driven d -wave pairing), sits at ≈ 133 K, and is itself less than half of room temperature. The honest reading of the empirical record (Fig. 2) is therefore twofold: the quantum-geometric flat-band route is not merely unproven for room temperature but bounded below it, and even the best-known ambient mechanism of any kind falls short. Our contribution is to convert the first half of that statement from a hunch into a ten-route closed-negative with an explicit mechanism for each closure.

What would reopen the axis. For completeness we state the conditions under which this closure would be falsified, so that the result is itself falsifiable. The axis reopens if any of the following is demonstrated: (i) a realizable ambient, dynamically-stable flat-band host with $\langle g \rangle \cdot U$ a factor $\gtrsim 5$ above the kagome/Lieb baseline and free of competing order; (ii) a beyond-mean-field pairing mechanism in which the pair-channel quantum metric provably exceeds the single-particle value (in direct contradiction to [2, 4, 5]); or (iii) a fully retarded-vertex sign-free quantum Monte Carlo with explicit phonons that exhibits a pair-channel $\langle g \rangle$ enhancement beyond the variational Lang–Firsov bound reported here. We regard (ii) as essentially excluded by the cited no-go results and (iii) as the single most informative remaining numerical experiment.

11 Limitations

The BKT T_c is a calibrated mean-field/BKT *estimate*, not a quantum Monte Carlo solve; we therefore anchor every quantitative claim to a measured reference and report the over-prediction band rather than a point value. The quantum metric is computed exactly, but in the overlap (Welch) normalization it is non-monotonic with stiffness for trivial localized bands, which is precisely why the Chern gate is mandatory and why dice/T_3 is excluded. Our closure is of the *quantum-geometric flat-band* axis at ambient pressure; it does not bound unconventional strong-correlation routes (cuprate/nickelate physics) beyond their measured ceilings, and it says nothing about high-pressure hydrides, which are excluded by the ambient criterion by construction. The one falsifier we could not execute in-house — a fully retarded-vertex sign-free quantum Monte Carlo with explicit phonons — is addressed here variationally and by published no-go results, not by a direct large-scale simulation.

12 Reproducibility

Every number in this paper is produced by a script and re-runnable without paid compute. The quantum-metric and BKT predictor is `tool/rtsc/fbgeom_predictor.py`: it computes $\langle g \rangle$ from tight-binding eigenvectors via the Fubini–Study link, applies the Fukui–Hatsugai–Suzuki Chern gate, and prints the deflated BKT band and the room-temperature deficit per host, alongside the `nick_*` substrate ranking of Section 8. The sign-free exact-diagonalization verification of Table 1 is `tool/rtsc/flatchern_meshconv.py` (two-body mesh-convergence) and `tool/rtsc/flatchern_sparse_mb.py` (many-body twisted-boundary D_s). The retarded-vertex closure of route 8 is `tool/rtsc/lang_firsov_offdiag_metric.py`. Each verdict is additionally recorded, with the verbatim computed numbers, in the project’s append-only ING ledger and in the architecture single-source-of-truth node `rtsc/answer_key_topdown`; no claim in the tier ledger (Table, Section 9) rests on a language-model self-judgement. The calibration anchors ($\langle g \rangle_{\text{ref}} = 0.672$, $(U/\Omega)_{\text{ref}} = 1.08$, the real-superconductor over-prediction band) are shared with all prior candidate verdicts in the campaign, so the estimator is consistent rather than re-fit per claim.

Scope of the released estimator. The predictor is intended for *ranking and gap-sizing* among topological flat-band hosts, not for absolute T_c prediction; it deliberately refuses to trust the T_c of trivial localized bands (dice/T_3) and prints an explicit caveat to that effect. The nickelate strain law is included only as a cross-host convenience and is labelled, in code and in this paper, as a published design law (not a discovery). This fencing is what allows the tool to be reused in future campaigns without re-importing a closed, red-ocean lever as if it were open.

13 Conclusion

The quantum-geometric flat-band prescription is real and verifiable: a high-Chern flat band carries an anomalously large, mesh-robust geometric pair stiffness. It is nonetheless insufficient for ambient room-temperature superconductivity. Across ten independent routes — five primary engineering axes, three subtler beyond-mean-field cracks, the single best ambient geometric host, and the top-down empirical lever — the route is closed-negative, with a consistent $\sim 5\times$ standing deficit in $\langle g \rangle U$. The structural reason is singular: the superfluid weight is bounded by the single-particle Fubini–Study metric, so no static or dynamical geometric mechanism multiplies it into the room-temperature regime, while the empirical ambient ceiling for *any* mechanism is the cuprate at less than half of room temperature. Ambient room-temperature superconductivity remains undiscovered and unconfirmed; this work removes one heavily-trafficked candidate axis from the search.

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